

Experiment 1: Class Fundamentals

Program 1.1: Student Information System

```
class Student {  
    int rollNo;  
    String name;  
    String branch;  
    double cgpa;  
    void display() {  
        System.out.println("Roll Number: " + rollNo);  
        System.out.println("Name: " + name);  
        System.out.println("Branch: " + branch);  
        System.out.println("CGPA: " + cgpa);  
    }  
}  
  
public class StudentInfo {  
    public static void main(String[] args) {  
        Student s = new Student();  
        s.rollNo = 101;  
        s.name = "Raina";  
        s.branch = "CSE-AIML";  
        s.cgpa = 9.2;  
        s.display();  
    }  
}
```

Output:

Roll Number: 101

Name: Raina

Branch: CSE-AIML

CGPA: 9.2

Program 1.2: Employee Information System

```
class Employee {  
    int employeeId;  
    String name;  
    String department;  
    double salary;  
    void display() {  
        System.out.println("Employee ID: " + employeeId);  
        System.out.println("Name: " + name);  
        System.out.println("Department: " + department);  
        System.out.println("Salary: " + salary);  
    }  
}  
  
public class EmployeeInfo {  
    public static void main(String[] args) {  
        Employee e = new Employee();  
        e.employeeId = 101;  
        e.name = "Raina";  
        e.department = "IT";  
        e.salary = 50000;  
        e.display();  
    }  
}
```

Output:

Employee ID: 101

Name: Raina

Department: IT

Salary: 50000.0

Experiment 2: Declaring Objects

Program 2.1: Book Details

```
class Book {  
    String title;  
    String author;  
    double price;  
    void display() {  
        System.out.println("Title: " + title);  
        System.out.println("Author: " + author);  
        System.out.println("Price: " + price);  
        System.out.println();  
    }  
}  
  
public class BookDetails {  
    public static void main(String[] args) {  
        Book b1 = new Book();  
        Book b2 = new Book();  
        Book b3 = new Book();  
        b1.title = "Java Programming";  
        b1.author = "James Gosling";  
        b1.price = 500;  
        b2.title = "Python Basics";  
        b2.author = "John Smith";  
        b2.price = 400;  
        b3.title = "Data Structures";  
        b3.author = "Mark Allen";  
        b3.price = 600;  
        b1.display();  
        b2.display();  
        b3.display();  
    }  
}
```

```
}  
}
```

Output:

Title: Java Programming

Author: James Gosling

Price: 500.0

Title: Python Basics

Author: John Smith

Price: 400.0

Title: Data Structures

Author: Mark Allen

Price: 600.0

Program 2.2: Array of Student Objects

```
import java.util.Scanner;  
  
class Student {  
    int rollNo;  
    String name;  
    String branch;  
    void display() {  
        System.out.println("Roll Number: " + rollNo);  
        System.out.println("Name: " + name);  
        System.out.println("Branch: " + branch);  
        System.out.println();  
    }  
}  
  
public class StudentArray {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        Student[] students = new Student[5];  
    }  
}
```

```
for (int i = 0; i < 5; i++) {  
    students[i] = new Student();  
    System.out.println("Enter details of Student " + (i + 1));  
    System.out.print("Roll Number: ");  
    students[i].rollNo = sc.nextInt();  
    sc.nextLine();  
    System.out.print("Name: ");  
    students[i].name = sc.nextLine();  
    System.out.print("Branch: ");  
    students[i].branch = sc.nextLine();  
}  
System.out.println("\nStudent Details:");  
for (int i = 0; i < 5; i++) {  
    students[i].display();  
}  
sc.close();  
}  
}
```

Output:

Enter details of Student 1

Roll Number: 101

Name: Raina

Branch: CSE-AIML

Enter details of Student 2

Roll Number: 102

Name: Rinku

Branch: CSE

Enter details of Student 3

Roll Number: 103

Name: Ananya

Branch: CSE-AIML

Enter details of Student 4

Roll Number: 104

Name: Kiran

Branch: ECE

Enter details of Student 5

Roll Number: 105

Name: Priya

Branch: IT

Student Details:

Roll Number: 101

Name: Raina

Branch: CSE-AIML

Roll Number: 102

Name: Rinku

Branch: CSE

Roll Number: 103

Name: Ananya

Branch: CSE-AIML

Roll Number: 104

Name: Kiran

Branch: ECE

Roll Number: 105

Name: Priya

Branch: IT

Experiment 3: Assigning Object Reference Variables

Program 3.1: Reference Assignment

```
class Student {
```

```
    String name;
```

```

        void display() {
            System.out.println("Name: " + name);
        }
    }

    public class ReferenceAssignment {
        public static void main(String[] args) {
            Student s1 = new Student();
            s1.name = "Raina";
            Student s2 = s1;
            System.out.println("Before modification:");
            s2.display();
            s1.name = "Rinku";
            System.out.println("After modification:");
            s2.display();
        }
    }

```

Output:

Before modification:

Name: Raina

After modification:

Name: Rinku

Program 3.2: Comparing Object References

```

class Employee {
    int id;
    String name;
    Employee(int id, String name) {
        this.id = id;
        this.name = name;
    }
}

```

```

}

public class CompareReferences {

    public static void main(String[] args) {

        Employee e1 = new Employee(101, "Rahul");

        Employee e2 = e1;

        if (e1 == e2) {

            System.out.println("Both references point to the same object.");

        } else {

            System.out.println("References point to different objects.");

        }

    }

}

```

Output:

Both references point to the same object.

Experiment 4: Introducing Methods

Program 4.1: Calculator Using Methods

```

class Calculator {

    int add(int a, int b) {

        return a + b;

    }

    int subtract(int a, int b) {

        return a - b;

    }

    int multiply(int a, int b) {

        return a * b;

    }

    double divide(int a, int b) {

        return (double) a / b;

    }

}

```

```

}

public class CalculatorDemo {

    public static void main(String[] args) {

        Calculator c = new Calculator();

        int a = 20;

        int b = 10;

        System.out.println("Addition: " + c.add(a, b));

        System.out.println("Subtraction: " + c.subtract(a, b));

        System.out.println("Multiplication: " + c.multiply(a, b));

        System.out.println("Division: " + c.divide(a, b));

    }

}

```

Output:

Addition: 30

Subtraction: 10

Multiplication: 200

Division: 2.0

Program 4.2: Rectangle Operations

```

class Rectangle {

    double length;

    double width;

    double area() {

        return length * width;

    }

    double perimeter() {

        return 2 * (length + width);

    }

    void display() {

        System.out.println("Area: " + area());

        System.out.println("Perimeter: " + perimeter());

    }

}

```